When does multivariate forecast reconciliation help?

Supplementary material

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Experiment 2

Table S1 compares separate and joint MinT reconciliation with the base forecasts in Experiment 2. In every scenario, reconciliation reduces MSE by 6–13% relative to the ARIMA base forecasts and by 1–5% relative to the VAR base forecasts. The relative MSEs of separate and joint reconciliation never differ by more than 0.019.

Table S1: Experiment 2: MSE of separate and joint MinT reconciliation relative to the base forecasts, averaged over 500 replications, 16 series and horizons 1–12.

		ARIMA		VAR	
Scenario	T	Separate	Joint	Separate	Joint
Separable, $\rho = 0.7$	108	0.941	0.938	0.964	0.966
Separable, $\rho = 0.7$	400	0.905	0.886	0.981	0.980
Separable, $\rho = -0.7$	108	0.939	0.936	0.979	0.983
Separable, $\rho = -0.7$	400	0.896	0.884	0.991	0.990
Variable-specific dynamics	108	0.924	0.930	0.973	0.974
Variable-specific dynamics	400	0.889	0.885	0.977	0.974
Node-varying correlation	108	0.936	0.933	0.958	0.963
Node-varying correlation	400	0.893	0.885	0.982	0.980
Variable-specific node covariance	108	0.944	0.942	0.977	0.980
Variable-specific node covariance	400	0.913	0.905	0.982	0.983
Node-varying dynamics	108	0.939	0.938	0.960	0.960
Node-varying dynamics	400	0.888	0.875	0.959	0.954

Application: ETS base forecasts

Table S2 repeats the accuracy comparison of the main text with base forecasts from automatically selected ETS models. For ETS we estimate the covariance matrices from response residuals, because the innovation residuals of multiplicative-error models are relative errors.

Table S2: Application: MSE relative to base forecasts (ETS), geometric mean over the series at each level, averaged over 84 forecast origins (73 with all twelve horizons) and horizons 1–12.

	Admissions			Dismissals		
Method	Total	Regions	States	Total	Regions	States
Base	1.000	1.000	1.000	1.000	1.000	1.000
OLS	0.997	1.003	1.310	0.985	1.070	1.288
WLS (structural)	0.990	0.928	1.120	0.943	0.984	1.122
MinT, separate	0.949	0.883	0.996	0.877	0.902	0.964
MinT, separate (joint estimate)	0.958	0.889	1.000	0.880	0.905	0.966
MinT, joint	0.949	0.878	0.988	0.815	0.852	0.925

Application: results by horizon

Table S3 breaks down the comparison of joint and separate reconciliation (ARIMA base forecasts) by forecast horizon. For admissions the ratio is above one at every horizon and level, and falls towards one as the horizon grows. For dismissals it is below one at every horizon and level.

Table S3: Application (ARIMA base forecasts): MSE of joint relative to separate MinT reconciliation by forecast horizon, geometric mean over the series at each level.

	Admissions			Dismissals		
Horizon	Total	Regions	States	Total	Regions	States
1–3	1.059	1.045	1.034	0.912	0.923	0.957
4–6	1.042	1.033	1.029	0.884	0.890	0.925
7–9	1.033	1.023	1.025	0.910	0.901	0.927
10–12	1.017	1.010	1.013	0.936	0.921	0.939

Application: results for every series

Table S4 gives the results series by series. For dismissals, joint reconciliation is better than separate reconciliation for every series. For admissions it is better for only 8 of the 33 series, and the differences are small.

Table S4: Application (ARIMA base forecasts): MSE of separate and joint MinT reconciliation relative to the base forecasts for every series, averaged over 84 origins (73 with all twelve horizons) and horizons 1–12.

	Admissions		Dismissals	
Series	Separate	Joint	Separate	Joint
Total	0.952	0.981	1.420	1.302
Midwest	1.017	1.019	0.879	0.802
Northeast	1.005	1.053	0.906	0.790
North	0.914	0.911	1.430	1.266
Southeast	0.998	1.034	1.249	1.176
South	0.975	1.000	0.890	0.839
AC	0.997	1.019	1.125	1.109
AL	0.944	0.960	0.959	0.933
AM	1.012	0.991	1.297	1.086
AP	0.957	0.911	1.145	1.100
BA	0.979	0.985	1.223	1.131
CE	0.962	1.017	1.061	0.929
DF	0.940	0.938	1.185	1.144
ES	0.984	0.994	1.066	0.992
GO	0.972	0.957	1.082	0.980
MA	0.929	0.963	1.176	1.111
MG	0.974	0.997	1.015	0.946
MS	0.948	0.909	1.250	1.150
MT	0.890	0.960	1.071	1.003
PA	0.923	0.931	1.141	1.098
PB	0.929	0.953	1.012	0.974
PE	0.939	0.962	1.111	0.985
PI	0.894	1.006	1.256	1.190
PR	0.999	1.025	1.194	1.155
RJ	0.994	1.026	1.058	1.019
RN	0.903	1.017	1.213	1.115
RO	0.973	0.973	1.160	1.021
RR	0.896	0.922	1.045	1.010
RS	0.999	1.054	1.083	1.012
SC	0.995	0.985	1.068	0.993
SE	0.943	0.934	1.086	1.075
SP	0.983	1.023	0.943	0.899
TO	0.837	0.864	1.244	1.166